

Can B. Kalayci, Ph.D.

Professor of Operations Research and Artificial Intelligence

Department of Industrial Engineering
Faculty of Engineering, Pamukkale University
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Online Academic Profiles

- **LinkedIn:** linkedin.com/in/cbkalayci
- **Google Scholar:** scholar.google.com/citations?user=iHetZQAAAAJ
- **ResearchGate:** researchgate.net/profile/Can-Kalayci
- **ORCID:** 0000-0003-2355-7015
- **Web of Science:** Researcher ID: 1057347
- **Scopus:** Author ID: 54951330900

Research Interests

- **Design and Analysis of Optimization and AI Algorithms:** Development, hybridization, and performance evaluation of metaheuristics and machine learning techniques for complex decision-making problems.
 - **Application Domains:** Production and disassembly line balancing, scheduling, electric vehicle routing, and portfolio optimization.
 - **Methodological Focus:** Combinatorial optimization, data-driven forecasting, environmentally conscious modeling, and sustainability-driven systems design.

Education

Northeastern University

Ph.D. in Industrial Engineering

Boston, MA, USA

2008–2012

- Dissertation: *“Algorithms for Solving Sequence-Dependent Disassembly Line Balancing Problems”*
- Supervisor: Prof. Surendra M. Gupta
- Available at: *Northeastern University Repository*

Pamukkale University

M.Sc. in Industrial Engineering

Denizli, Turkey

2005–2008

- Thesis: *“A Student Success-Oriented Examination Timetabling Model and Its Software Implementation”*
- Supervisor: Prof. Askiner Gungor

Sakarya University

B.Sc. in Computer Engineering

Sakarya, Turkey

2001–2005

- Graduated with High Honors

Selected Publications

- [1] M. A. Akbay, C. Blum, and C. B. Kalayci, "Cmsa based on set covering models for packing and routing problems", *Annals of Operations Research*, vol. 343, no. 1, pp. 1–38, 2024.
- [2] J. Alcazar, M. Ghazi Vakili, C. Kalayci, and A. Perdomo-Ortiz, "Enhancing combinatorial optimization with classical and quantum generative models", *Nature Communications*, vol. 15, no. 1, 2024.
- [3] E. Güler, C. Kalayci, M. Ali Ilgin, E. Özceylan, and A. Güngör, "Advances in partial disassembly line balancing: A state-of-the-art review", *Computers and Industrial Engineering*, vol. 188, 2024.
- [4] M. Akbay, C. Kalayci, C. Blum, and O. Polat, "Variable Neighborhood Search for the Two-Echelon Electric Vehicle Routing Problem with Time Windows", *Applied Sciences (Switzerland)*, vol. 12, no. 3, 2022.
- [5] O. Polat, C. Berk Kalayci, and D. Topaloğlu, "Modelling and solving the milk collection problem with realistic constraints", *Computers and Operations Research*, vol. 142, 2022.
- [6] M. Soyer, C. Kalayci, and Ö. Karakaş, "Low-cycle fatigue parameters and fatigue life estimation of high-strength steels with artificial neural networks", *Fatigue and Fracture of Engineering Materials and Structures*, vol. 45, no. 12, pp. 3764–3785, 2022.
- [7] Y. Yilmaz and C. Kalayci, "Variable Neighborhood Search Algorithms to Solve the Electric Vehicle Routing Problem with Simultaneous Pickup and Delivery", *Mathematics*, vol. 10, no. 17, 2022.
- [8] M. Akbay and C. Kalayci, "A Variable Neighborhood Search Algorithm for Cost-Balanced Travelling Salesman Problem", presented at the Advances in Intelligent Systems and Computing, vol. 1332 AIST, 2021, pp. 23–36.
- [9] O. Polat, S. Akçok, M. Akbay, D. Topaloğlu, S. Arslan, and C. Kalayci, "Classification of raw cow milk using information fusion framework", *Journal of Food Measurement and Characterization*, vol. 15, no. 6, pp. 5113–5130, 2021.
- [10] M. Akbay, C. Kalayci, and O. Polat, "A parallel variable neighborhood search algorithm with quadratic programming for cardinality constrained portfolio optimization", *Knowledge-Based Systems*, vol. 198, 2020.
- [11] S. Gupta, A. Güngör, K. Govindan, E. Özceylan, C. Kalayci, and R. Piplani, "Responsible & sustainable manufacturing", *International Journal of Production Research*, vol. 58, no. 23, pp. 7181–7182, 2020.
- [12] C. Kalayci, S. Karagoz, and Ö. Karakas, "Soft computing methods for fatigue life estimation: A review of the current state and future trends", *Fatigue and Fracture of Engineering Materials and Structures*, vol. 43, no. 12, pp. 2763–2785, 2020.
- [13] C. Kalayci, O. Polat, and M. Akbay, "An efficient hybrid metaheuristic algorithm for cardinality constrained portfolio optimization", *Swarm and Evolutionary Computation*, vol. 54, 2020.
- [14] C. Kalayci, O. Ertenlice, and M. Akbay, "A comprehensive review of deterministic models and applications for mean-variance portfolio optimization", *Expert Systems with Applications*, vol. 125, pp. 345–368, 2019.
- [15] C. Kalayci, S. Karagöz, and Ö. Karakaş, "Bee colony intelligence in fatigue life estimation of simulated magnesium alloy welds", *International Journal of Fatigue*, vol. 127, pp. 36–44, 2019.
- [16] E. Özceylan, C. Kalayci, A. Güngör, and S. Gupta, "Disassembly line balancing problem: A review of the state of the art and future directions", *International Journal of Production Research*, vol. 57, no. 15-16, pp. 4805–4827, 2019.
- [17] O. Ertenlice and C. Kalayci, "A survey of swarm intelligence for portfolio optimization: Algorithms and applications", *Swarm and Evolutionary Computation*, vol. 39, pp. 36–52, 2018.
- [18] C. Kalayci, O. Ertenlice, H. Akyer, and H. Aygoren, "An artificial bee colony algorithm with feasibility enforcement and infeasibility toleration procedures for cardinality constrained portfolio optimization", *Expert Systems with Applications*, vol. 85, pp. 61–75, 2017.

- [19] C. Kalayci and C. Kaya, "An ant colony system empowered variable neighborhood search algorithm for the vehicle routing problem with simultaneous pickup and delivery", *Expert Systems with Applications*, vol. 66, pp. 163–175, 2016.
- [20] C. Kalayci, O. Polat, and S. Gupta, "A hybrid genetic algorithm for sequence-dependent disassembly line balancing problem", *Annals of Operations Research*, vol. 242, no. 2, pp. 321–354, 2016.
- [21] O. Polat, C. Kalayci, O. Mutlu, and S. Gupta, "A two-phase variable neighbourhood search algorithm for assembly line worker assignment and balancing problem type-II: An industrial case study", *International Journal of Production Research*, vol. 54, no. 3, pp. 722–741, 2016.
- [22] C. Kalayci, A. Hancilar, A. Gungor, and S. Gupta, "Multi-objective fuzzy disassembly line balancing using a hybrid discrete artificial bee colony algorithm", *Journal of Manufacturing Systems*, vol. 37, pp. 672–682, 2015.
- [23] C. Kalayci, O. Polat, and S. Gupta, "A variable neighbourhood search algorithm for disassembly lines", *Journal of Manufacturing Technology Management*, vol. 26, no. 2, pp. 182–194, 2015.
- [24] O. Polat, C. Kalayci, O. Kulak, and H.-O. Günther, "A perturbation based variable neighborhood search heuristic for solving the Vehicle Routing Problem with Simultaneous Pickup and Delivery with Time Limit", *European Journal of Operational Research*, vol. 242, no. 2, pp. 369–382, 2015.
- [25] C. Kalayci and S. Gupta, "A tabu search algorithm for balancing a sequence-dependent disassembly line", *Production Planning and Control*, vol. 25, no. 2, pp. 149–160, 2014.
- [26] C. Kalayci and S. Gupta, "A particle swarm optimization algorithm with neighborhood-based mutation for sequence-dependent disassembly line balancing problem", *International Journal of Advanced Manufacturing Technology*, vol. 69, no. 1-4, pp. 197–209, 2013.
- [27] C. Kalayci and S. Gupta, "Ant colony optimization for sequence-dependent disassembly line balancing problem", *Journal of Manufacturing Technology Management*, vol. 24, no. 3, pp. 413–427, 2013.
- [28] C. Kalayci and S. Gupta, "Artificial bee colony algorithm for solving sequence-dependent disassembly line balancing problem", *Expert Systems with Applications*, vol. 40, no. 18, pp. 7231–7241, 2013.
- [29] C. Kalayci and S. Gupta, "Balancing a sequencedependent disassembly line using simulated annealing algorithm", *Applications of Management Science*, vol. 16, pp. 81–103, 2013.
- [30] C. Kalayci and A. Güngör, "A genetic algorithm based examination timetabling model focusing on student success for the case of the college of engineering at Pamukkale University, Turkey", *Gazi University Journal of Science*, vol. 25, no. 1, pp. 137–153, 2012.

Research Projects

- **New Models and Solution Approaches for the Sustainable Partial Disassembly Line Balancing Problem**
TÜBİTAK 1001 Grant, Turkey 2024–Ongoing
 - Principal Investigator of this 3-year national research project, focused on the development of environmentally sustainable disassembly line models integrating partial disassembly dynamics and optimization techniques.
- **Developing Efficient Routing Algorithms for Sustainable City Logistics**
TÜBİTAK 1001 Grant, Turkey 2019–2022
 - Led the project to design metaheuristic-based routing strategies supporting eco-friendly and cost-efficient urban logistics systems.
- **An Integrated Model for the Milk Collection Problem and Development of Efficient Solution Approaches**
TÜBİTAK 3501 Grant, Turkey 2018–2020
 - Served as project researcher. The project developed novel mathematical models and heuristics for perishable goods collection under routing constraints. Project PI: Dr. Olcay Polat.
- **Determination of Fatigue Behaviors of Micro-Structure Magnesium Alloy Using Heuristic Algorithms**
Pamukkale University Internal Grant, Turkey 2018–2019
 - Principal Investigator. Applied optimization algorithms to model material fatigue under varying microstructural configurations in lightweight alloys.
- **Designing Efficient Algorithms for Cardinality-Constrained Portfolio Optimization**
TÜBİTAK 3501 Grant, Turkey 2015–2017
 - Principal Investigator. Designed and implemented hybrid heuristic algorithms to solve large-scale portfolio selection problems under cardinality and risk constraints.
- **Developing Models and Solution Approaches for Vehicle Routing Problems**
Pamukkale University Internal Grant, Turkey 2015–2016
 - Principal Investigator. Focused on variants of the vehicle routing problem (VRP) including time windows and service level constraints using advanced metaheuristics.
- **Solution Approaches for the Feeder Containership Routing Problem**
Pamukkale University Internal Grant, Turkey 2014–2015
 - Project researcher. Contributed to the modeling and optimization of maritime feeder containership routes. Project PI: Dr. Olcay Polat.
- **Solving Vehicle Routing Problems with Simultaneous Pickup and Delivery under Time Limits Using Metaheuristics**
Pamukkale University Internal Grant, Turkey 2014–2015
 - Principal Investigator. Addressed complex VRP variants involving bidirectional logistics and time constraints through customized heuristic methods.

Academic and Professional Experience

Department of Industrial Engineering, Faculty of Engineering, Pamukkale University
TÜBİTAK Project Evaluator and Panelist

Denizli, TR
2015–Ongoing

- Served as a panelist and external referee for over 100 research proposals submitted to various TÜBİTAK funding schemes.
- Contributed to detailed scientific evaluation reports across domains including AI, optimization, and sustainable systems.
- Authored summary reports, reviewer syntheses, and funding recommendations.

Department of Industrial Engineering, Faculty of Engineering, Pamukkale University
Erasmus+ Departmental Coordinator

Denizli, TR
2018–Ongoing

- Managed Erasmus+ mobility programs for students and staff, overseeing bilateral agreements and academic recognition.
- Provided academic counseling for incoming and outgoing students, including course equivalency and transcript management.
- Assisted in the preparation and submission of EU project proposals in cooperation with partner institutions.

Faculty of Engineering, Pamukkale University
Associate Dean

Denizli, TR
2021–2022

- Led academic planning and quality assurance processes across multiple departments.
- Facilitated interdisciplinary collaboration and supported preparation of externally funded research proposals.
- Coordinated faculty development and performance evaluation processes.

Zapata Computing, Inc.
Research Consultant

Boston, MA
2021–2022

- Collaborated on the development of quantum-inspired algorithms for combinatorial optimization.
- Provided academic expertise on integrating metaheuristics with quantum computational frameworks.

College of Engineering, Northeastern University
Teaching Assistant

Boston, MA
2010–2011

- Maintained computing laboratories and supported instruction in optimization and algorithmic courses.
- Assisted students in both undergraduate and graduate-level projects and assignments.

Cybernetics and AI Lab, Technical University of Košice
Research Intern

Košice, Slovakia
Summer 2003

- Developed software components for intelligent systems as part of a European student research program.
- Engaged in prototyping of early agent-based simulation tools for dynamic systems.

Teaching Experience

- **ENM 543 – Artificial Intelligence Methodologies in Industrial Engineering**
Graduate course on machine learning, metaheuristics, and hybrid AI systems with applications in industrial problem solving.
- **IENG 435 – Heuristic Methods in Problem Solving**
Senior undergraduate course on the design and implementation of heuristic algorithms for NP-hard problems.
- **CENG 280 – Programming Basics and Algorithm Design**
Undergraduate course introducing fundamental programming concepts and algorithm development using Python.
- **CENG 191 – Introduction to Programming**
Introductory undergraduate course focused on procedural programming and logical structures.
- **CENG 110 – Intermediate Programming**
Continuation of CENG 191, emphasizing object-oriented programming concepts and algorithmic problem solving.
- **ENM 555 – MATLAB and Applications**
Graduate-level course covering numerical computing, modeling, and algorithm development using MATLAB.
- **IENG 473 – Portfolio Optimization**
Elective undergraduate course exploring quantitative finance, risk models, and solution techniques including cardinality constraints.
- **ENM 503 – Statistical Experimental Design**
Graduate course covering Taguchi methods, factorial designs, and response surface methodology.
- **IENG 217 – Probability and Statistics**
Core undergraduate course introducing foundational statistical analysis and probability theory.
- **IENG 426 – Environmentally Conscious Manufacturing**
Elective undergraduate course addressing sustainable manufacturing, green metrics, and life-cycle assessment.
- **MUHY 578 – Management Information Systems**
Graduate-level interdisciplinary course on the role of information systems in enterprise decision-making.
- **IENG 108 – Industrial Engineering Applications**
First-year orientation course introducing students to the scope and case studies of industrial engineering.
- **MUHF 401 – Case Studies in Engineering**
Capstone course focused on interdisciplinary engineering problems and real-world case analysis.

Supervised Theses

- **Yusuf Yılmaz**, Ph.D., Pamukkale University, 2023
Title: *“Variable Neighborhood Search Algorithms for Solving Electric Vehicle Routing Problems”*
Role: Primary Advisor
- **Mehmet Anıl Akbay**, M.Sc., Pamukkale University, 2019
Title: *“A Solution Approach Based on Variable Neighborhood Search Algorithm for Cardinality-Constrained Portfolio Optimization”*
Role: Primary Advisor
- **Umut Mete**, M.Sc., Pamukkale University, 2019
Title: *“A Variable Neighborhood Search Approach for Permutation Flow-Shop Scheduling”*
Role: Primary Advisor
- **Sevcan Karagöz**, M.Sc., Pamukkale University, 2019
Title: *“Determination of Fatigue Behaviors of Micro-Structure Magnesium Alloy Using Heuristic Algorithms”*
Role: Primary Advisor
- **Ökkeş Ertenlice**, M.Sc., Pamukkale University, 2018
Title: *“An Artificial Bee Colony Algorithm-Based Solution Approach for Cardinality-Constrained Portfolio Optimization”*
Role: Primary Advisor

- **Can Kaya**, M.Sc., Pamukkale University, 2017
Title: “An Ant Colony System Empowered Variable Neighborhood Search Algorithm for the Vehicle Routing Problem with Simultaneous Pickup and Delivery”
Role: Primary Advisor
- **Hasan Akyer**, Ph.D., Pamukkale University, 2016
Title: “Heuristic Approaches for Optimal Portfolio Management”
Role: Co-Advisor

Scholarships and Awards

- Included in the *World’s Top 2% of Scientists* list compiled by Stanford University and published by Elsevier, based on standardized citation indicators (2021–2024).
- Citation Award recipient at the Pamukkale University Annual Prize-Giving Ceremony for exceptional research impact (2019).
- Bronze Medal, Metaheuristics Competition at the International Metaheuristics Summer School (MESS), Italy; recognized among top global participants (2018).
- Recipient of a fully funded Ph.D. scholarship from the Council of Higher Education (YÖK), Turkey; awarded competitively for doctoral studies abroad (2008–2012).
- Graduated with High Honors, Sakarya University, B.Sc. in Computer Engineering; recognized for academic excellence (2005).

Technical Skills

- **Programming:** Python, MATLAB, C/C++, Julia
- **Optimization:** Gurobi, IBM CPLEX
- **Data Analysis & ML:** pandas, NumPy, Scikit-learn
- **Scientific Writing:** LaTeX, Overleaf

Languages

- **English:** C1–C2 (Proficient)
- **Turkish:** Native speaker (C2)

Editorial and Peer Review Activities

Elsevier Journals

- *European Journal of Operational Research*
- *Computers & Operations Research*
- *Computers & Industrial Engineering*
- *Journal of Cleaner Production*
- *Journal of Manufacturing Systems*
- *Applied Soft Computing*
- *Knowledge-Based Systems*
- *Swarm and Evolutionary Computation*
- *Expert Systems with Applications*
- *International Journal of Production Economics*

- *Engineering Applications of Artificial Intelligence*
- *Transportation Research Part B: Methodological*

Springer Journals

- *Annals of Operations Research*
- *Financial Innovation*

Taylor & Francis Journals

- *International Journal of Production Research*

Emerald Journals

- *Journal of Manufacturing Technology Management*